Feature Report: Velocity Features

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Basic Information

- **Total Customers**: 61,410
- **Total Features**: 16
- **Feature Columns**: txn_count_total, txn_per_day_avg, txn_velocity, high_frequency_customer_flag, volume_debit_total, volume_credit_total, account_turnover_rate, flow_through_velocity_hours, time_between_deposit_withdrawal, sudden_inflow_outflow_pattern, txn_count_eft, volume_eft_total, txn_velocity_eft, volume_eft_sudden_increase, txn_count_eft_spike, velocity_change_eft

Data Quality Checks

☑ No Null Values

All features have complete data.

☑ No Duplicate Customer IDs

Feature Statistics

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
`txn_count_total`	Integer	1	4,991	96.13	63	102.92	0	0.00%
`txn_per_day_avg`	Float	0.00	393.00	1.19	0.72	3.97	907	1.48%
`txn_velocity`	Float	0.00	393.00	1.19	0.72	3.97	907	1.48%
`high_frequency_customer_flag`	Integer	0	1	0.01	0	0.08	61,045	99.41%
`volume_debit_total`	Float	0.00	1437637276.00	2861381.83	886504.50	14983096.02	550	0.90%
`volume_credit_total`	Float	0.00	2052481891.00	3259518.81	1213366.50	18248172.39	2,080	3.39%
`account_turnover_rate`	Float	0.00	10.00	1.87	0.77	2.73	550	0.90%
`flow_through_velocity_hours`	Float	0.00	2184.00	57.24	28.17	108.68	4,391	7.15%
`time_between_deposit_withdrawal`	Float	0.00	2184.00	57.24	28.17	108.68	4,391	7.15%
`sudden_inflow_outflow_pattern`	Integer	0	1	0.79	1	0.41	13,171	21.45%
`txn_count_eft`	Integer	0	4,959	17.44	11	43.14	10,863	17.69%
`volume_eft_total`	Float	0.00	2675109760.00	2750979.67	790347.50	21009517.50	10,863	17.69%
`txn_velocity_eft`	Float	0.00	53.35	0.24	0.16	0.48	10,863	17.69%
`volume_eft_sudden_increase`	Integer	0	1	0.16	0	0.37	51,582	84.00%
`txn_count_eft_spike`	Integer	0	0	0.00	0	0.00	61,410	100.00%
`velocity_change_eft`	Integer	0	0	0.00	0	0.00	61,410	100.00%

Validation Checks

☑ No Unexpected Negative Values

☑ No Infinite Values

Outlier Detection (Values > 3 Standard Deviations)

- `txn_count_total`: 924 outliers (1.50%)
- `txn_per_day_avg`: 145 outliers (0.24%)

- `txn_velocity`: 145 outliers (0.24%)
- `high_frequency_customer_flag`: 365 outliers (0.59%)
- `volume_debit_total`: 544 outliers (0.89%)
- `volume_credit_total`: 363 outliers (0.59%)
- `flow_through_velocity_hours`: 1,009 outliers (1.64%)
- `time_between_deposit_withdrawal`: 1,009 outliers (1.64%)
- `txn_count_eft`: 380 outliers (0.62%)
- `volume_eft_total`: 326 outliers (0.53%)

... and 1 more features with outliers

Flag Feature Distributions

Feature	True Count	True Percentage	False Count	False Percentage
`high_frequency_customer_flag`	365	0.59%	61,045	99.41%

Summary

- ☑ **Total Features**: 16
- ☑ **Data Completeness**: 6.25%
- ☑ **Customers Covered**: 61,410
- ☑ **Flag Features**: 1

Report generated automatically by feature engineering pipeline